PROJECT DEVELOPMENT PHASE

Business Questions • Tableau Visualisations • Feature Demonstration • Website Integration

Student / Project Information

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Course	Bachelor of Computer Application (BCA)	Semester	Second Year / 3rd Semester
College	MEDI-CAPS University	Project Type	Individual Project

Global Fuel Prices and Taxation Trends: A Comprehensive Global Performance Intelligence Report

Purpose: To learn the concepts of Tableau and data analytics.

1. Business Questions & Visualisation Report

The project uses a historical fuel-pricing and taxation dataset to support descriptive comparisons across countries, regions, subsidy levels and income levels. The three dashboards are designed around policy analysis, supply-chain planning, and economic-development exploration.

Business question	Visualisation / purpose	Dashboard
Which countries have the highest average fuel-tax percentages?	Horizontal bar chart ranking countries by average tax percentage.	Fuel Taxation & Subsidy Policy
How does average petrol price vary across regions?	Regional bar chart with a region legend.	Fuel Taxation & Subsidy Policy
What relationship can be explored between taxation and petrol prices?	Scatter plot to compare association; it does not prove causation.	Fuel Taxation & Subsidy Policy
How do average fuel prices differ by subsidy level?	Grouped bar chart comparing diesel, LPG and petrol across subsidy levels.	Fuel Taxation & Subsidy Policy
Which countries are in the top 10 for average fuel price, and how do petrol and diesel compare?	Ranked stacked bar comparison of petrol and diesel.	Global Fuel Prices & Supply Chain Planning
Which countries are in the bottom 10 for average fuel price, and how do petrol and diesel compare?	Ranked stacked bar comparison of petrol and diesel.	Global Fuel Prices & Supply Chain Planning
How have Brent crude and fuel prices changed over time in the dataset?	Time-series line chart for Brent crude and regional petrol-price trends.	Global Fuel Prices & Supply Chain Planning
How do average petrol prices and tax percentages compare across income levels?	Bar charts for fuel prices and average tax percentage by income level.	Fuel Prices, Taxation & Economic Development

Scope note: The charts describe patterns in the available CSV data (January 2020–April 2026).
They do not provide live market prices or establish cause-and-effect relationships.

2. Dashboard Development & Feature Demonstration

Core implemented features documented without embedding dashboard screenshots

The project includes three scenario dashboards and supporting visualisation, publishing, and website-integration features. The table below summarises the implemented components without claiming unverified performance testing.

Feature	Status	What it provides / limitation

Scenario 1 — Fuel Taxation & Subsidy Policy	Implemented; demonstrated	Country tax rankings, petrol prices by region, taxation-versus-petrol scatter plot, and fuel prices by subsidy level. Scatter plot indicates association only.
Scenario 2 — Global Fuel Prices & Supply Chain Planning	Implemented; demonstrated	Top/bottom country comparisons, petrol/diesel comparison, regional petrol prices, and historical Brent crude/fuel-price trends.
Scenario 3 — Fuel Prices, Taxation & Economic Development	Implemented; demonstrated	Income-level comparisons for fuel prices and tax percentages, with Income Level, Region and Subsidy Level controls visible in the workbook.
Visual comparison and legends	Implemented; demonstrated	Charts and legends support comparisons across countries, regions, fuel measures, tax percentages and subsidy categories. Interpret legends using the labels shown on each dashboard.
Published Tableau Public workbook	Published; demonstrated	The workbook is published online so viewers can open the dashboards through a browser and navigate the available scenario views.
Website integration and deployment	Implemented; demonstrated	The Flask-based website integrates the published Tableau visualisation and is deployed on Render, providing a public access point for the project.

Feature implementation summary

Tracked items	Implemented	Demonstrated	Recorded rate
6	6	6	100% (6 of 6 checklist items)

Counting note: the 100% figure is a simple documentation count for the six listed implemented features, not a measured delivery, usability, or platform-performance score.

3. Website Development & Integration

Project stack, public access, and development notes

A web interface complements the Tableau workbook by providing a single access point for the project overview and interactive dashboards. The project uses a lightweight Flask-based application with the published Tableau visualisation and online deployment.

Technology	Role in the project
Python	Programming language used for the web application.
Flask	Python web framework used to serve the website.
HTML and CSS	Structure and visual styling of website content.
Tableau Public	Hosts the published interactive visualisations used by the project.
GitHub	Stores the website source code and supporting project files in the repository.
Render	Deployment and online hosting platform for the web application.

Project access

Resource	Access
Tableau Public Story	Open the published Tableau Story
Live project website	fuel-prices-taxation-dashboard.onrender.com
GitHub repository	Harshvardhan-30/global-fuel-prices-taxation-analysis

Demo video	https://drive.google.com/file/d/1GLqMmXUegjqNgqAs9VH-QL4tpPIfsRss/view?usp=sharing

Development notes & limitations

- The dataset was connected from a CSV file directly in Tableau Public; a separate SQL database is not claimed.
- The dashboards support descriptive comparisons. Correlation or visual association should not be interpreted as causation.
- No live data feed, formal user study, load test, or mobile responsiveness test is claimed as independently verified.
- The project is an individual implementation; GitHub, Tableau Public, and the deployed website provide the main public project resources.

PROJECT PURPOSE

To learn the concepts of Tableau and data analytics by building scenario-based visualisations of global fuel prices, taxation and subsidy patterns.